

Pei-Chi Hu

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EDUCATION

Carnegie Mellon University (CMU), Pittsburgh, PA

Aug. 2026 — Dec. 2027

Master in Computer Vision

- Relevant Coursework: Advanced Computer Vision, Learning for 3D, Intro to Robot Learning

National Yang Ming Chiao Tung University (NYCU), Hsinchu, Taiwan

Sept. 2018 — June 2022

Bachelor of Science in Mechanical Engineering, Double Major in Electrical Engineering

SKILLS

- Deep Learning & Vision: Deep Neural Networks, PyTorch, OpenCV, 3D Perception, Computer Vision
- Programming & Infrastructure: C, C++, Python, JavaScript, Docker, Linux, Bash/Shell Scripting, Git, ROS

WORK EXPERIENCE

R&D Senior Software Engineer, Autonomous Intelligence Systems

Apr. 2025 — Jun. 2026

FARobot, Inc. (Joint Venture Foxconn and ADLINK)

Taipei, Taiwan

- Engineered a unified ROS2 map architecture using Thin-Plate Spline (TPS) to correct non-rigid deformations across heterogeneous fleets (SMR, MiR, HiK, SEER), reducing average alignment error by 44.2% along execution paths.
- Architected an automated camera-to-robot calibration system by formulating a 12-AprilTag layout and applying Singular Value Decomposition (SVD) to solve rigid body transformations, ensuring sub-millimeter spatial accuracy and eliminating manual cumulative errors.
- Integrated Intel RealSense depth cameras into ROS2 Navigation to expand obstacle detection into 3D space. Deeply analyzed and refactored core Voxel Layer source code to overcome native 16-voxel height limits, doubling vertical Fov representation and enabling robust detection of low-lying and overhead hazards.
- Developed a browser-triggerable RESTful API and PLC-integrated localization recovery pipeline utilizing AprilTag-to-URDF pose computation to instantly re-initialize navigation stacks upon tracking loss.

R&D Junior Software Engineer, Autonomous Intelligence Systems

Dec. 2022 — Apr. 2025

FARobot, Inc. (Joint Venture Foxconn and ADLINK)

Taipei, Taiwan

- Containerized multi-brand LiDAR and IMU drivers using Docker on Linux platforms, accelerating deployment time by 30% and ensuring cross-platform reproducibility.
- Optimized Extended Kalman Filter (EKF) parameters to prioritize high-fidelity sensor measurements over odometry prediction models, boosting stop-phase localization stability from 45.16% to 74.19%.
- Mitigated Cartographer mapping drift in 200m×200m feature-sparse corridors by re-engineering graph information matrices to de-emphasize uncertain local constraints.

PROJECTS

Robot Learning & Simulation Validation | Carnegie Mellon University

- Training and evaluating neural network control policies in simulated environments using PyTorch, aligning with closed-loop validation frameworks for deep learning.
- Overcame distribution shift in autonomous continuous control by developing a DAGger pipeline, iteratively refining the policy through expert demonstration querying to ensure stability.

AMR Digital Twin & Physics Simulation | Research Assistant

- Defined embedded hardware specifications via power budget analysis and engineered real-time STM32 firmware for closed-loop motor control and synchronized 4-wheel encoder updates.
- Developed a high-fidelity Gazebo digital twin to bridge the sim-to-real gap, ensuring robust control validation prior to hardware deployment.

Multi-shuttle ASRS with Quick Response Code | Lead Developer

- Architected a multi-shuttle control system driven by Python multiprocessing and mutex synchronization, efficiently coordinating 8 conveyors, 2 elevators, and 3 shuttles.
- Designed an integrated Warehouse Management System (WMS) featuring real-time web monitoring and QR code tracking via OpenCV, Django, and MySQL databases, published in the 39th CSME National Conference.